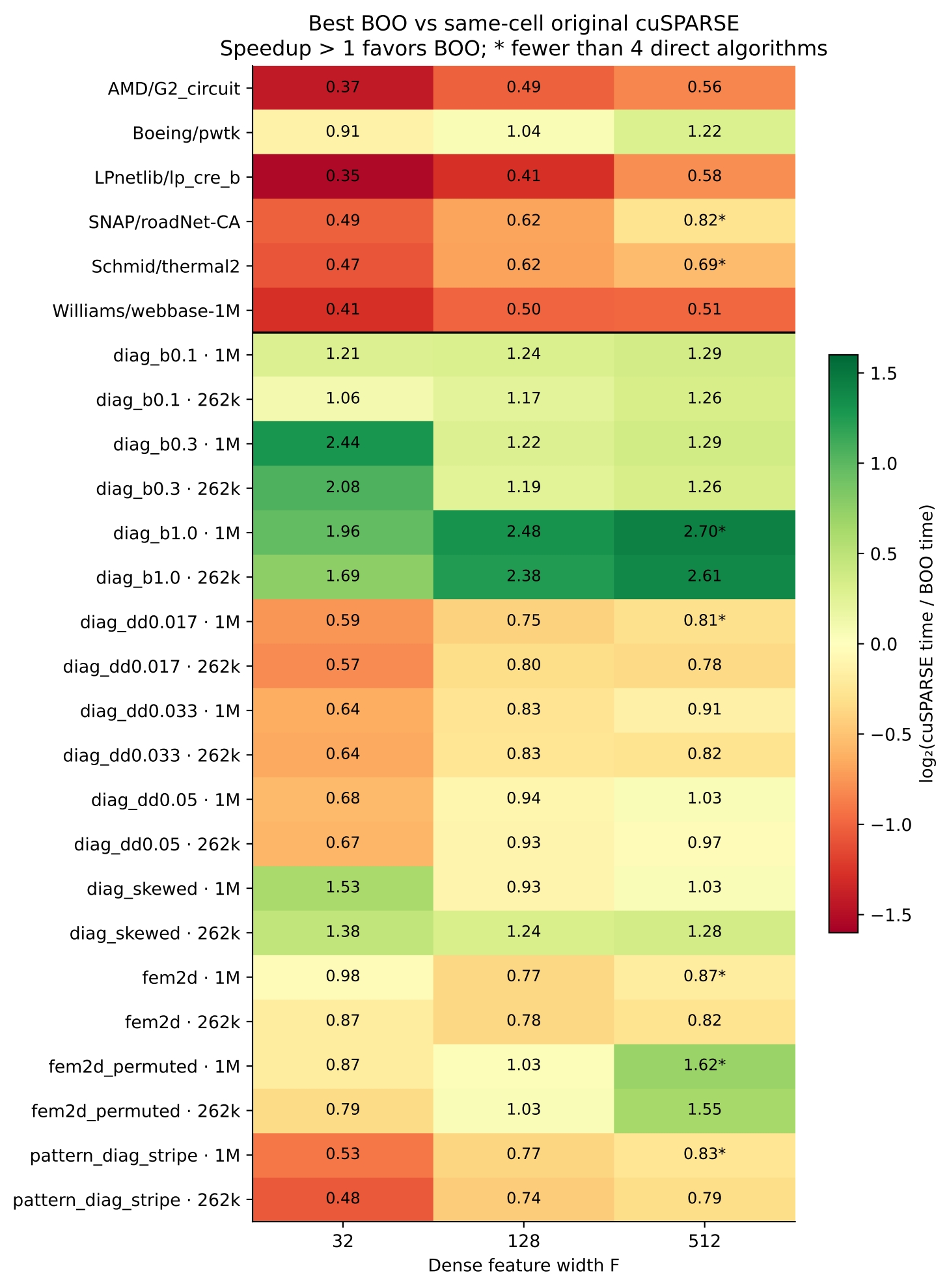
Best BOO vs same-cell original cuSPARSE
Speedup > 1 favors BOO; * fewer than 4 direct algorithms
AMD/G2_circuit 0.37 0.49 0.56
Boeing/pwtk 0.91 1.04 1.22
LPnetlib/lp_cre_b 0.35 0.41 0.58
SNAP/roadNet-CA 0.49 0.62 0.82*
Schmid/thermal2 0.47 0.62 0.69*
Williams/webbase-1M 0.41 0.50 0.51
diag_b0.1 · 1M 1.21 1.24 1.29
diag_b0.1 · 262k 1.06 1.17 1.26
diag_b0.3 · 1M 2.44 1.22 1.29
diag_b0.3 · 262k 2.08 1.19 1.26
diag_b1.0 · 1M 1.96 2.48 2.70*
diag_b1.0 · 262k 1.69 2.38 2.61
diag_dd0.017 · 1M 0.59 0.75 0.81*
diag_dd0.017 · 262k 0.57 0.80 0.78
diag_dd0.033 · 1M 0.64 0.83 0.91
diag_dd0.033 · 262k 0.64 0.83 0.82
diag_dd0.05 · 1M 0.68 0.94 1.03
diag_dd0.05 · 262k 0.67 0.93 0.97
diag_skewed · 1M 1.53 0.93 1.03
diag_skewed · 262k 1.38 1.24 1.28
fem2d · 1M 0.98 0.77 0.87*
fem2d · 262k 0.87 0.78 0.82
fem2d_permuted · 1M 0.87 1.03 1.62*
fem2d_permuted · 262k 0.79 1.03 1.55
pattern_diag_stripe · 1M 0.53 0.77 0.83*
pattern_diag_stripe · 262k 0.48 0.74 0.79
32
128
512
Dense feature width F
log₂(cuSPARSE time / BOO time)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5